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2026 Annual Symposium

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Causal AI and the Shift from Prediction to Action

AMIA 2026 KDDM Working Group Collaborative Workshop

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Disclosure of Relevant Financial Relationships

The following presenter(s) **have no** relevant financial relationship(s) with ineligible companies to disclose.

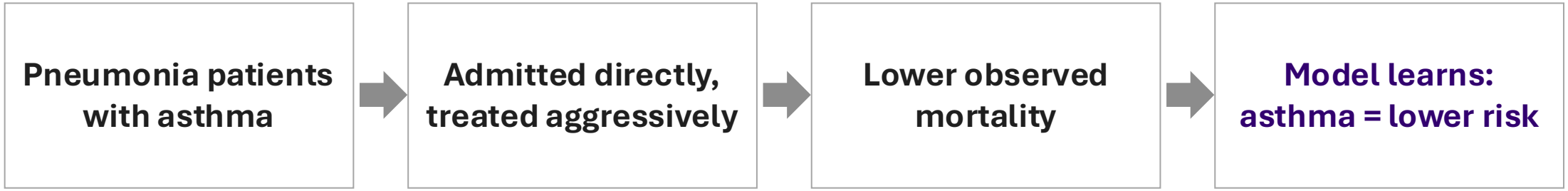
- Yi Guo, PhD

Learning Objectives

At the conclusion of this presentation the learner will be able to:

1. Distinguish predictive validity from decision-support validity in clinical AI
2. Assess whether electronic health record data can support the design components of a target trial emulation
3. Identify how deploying AI in clinical care creates feedback loops that alter the data used to update and evaluate models

An accurate model can recommend the wrong action



Used to decide who can be treated as an outpatient, the model could recommend discharging some of the highest-risk patients. The prediction fit the data; the implied action could cause harm.

The model learned outcomes under the care that was delivered. **Prediction asks "what will be observed next?" Action asks "what would happen if we chose differently?"**

Cooper et al. Artif Intell Med 1997; Caruana et al. KDD 2015

A good prediction is not necessarily a good decision

PREDICTIVE QUESTION

Who is likely to develop an obesity-associated cancer?

Estimates risk under the care patients actually received

CAUSAL QUESTION

Would starting a GLP-1 receptor agonist change that risk?

Compares outcomes under alternative treatment strategies

Clinicians make the second kind of decision every day. Most clinical AI answers the first. We will come back to this question.

The data-generating process is part of the causal problem

Clinical data are records of care decisions, not just of disease.

Who enters

Patients appear when they seek care, not when risk begins

What is recorded

Tests, notes, and codes follow clinical concern and workflow

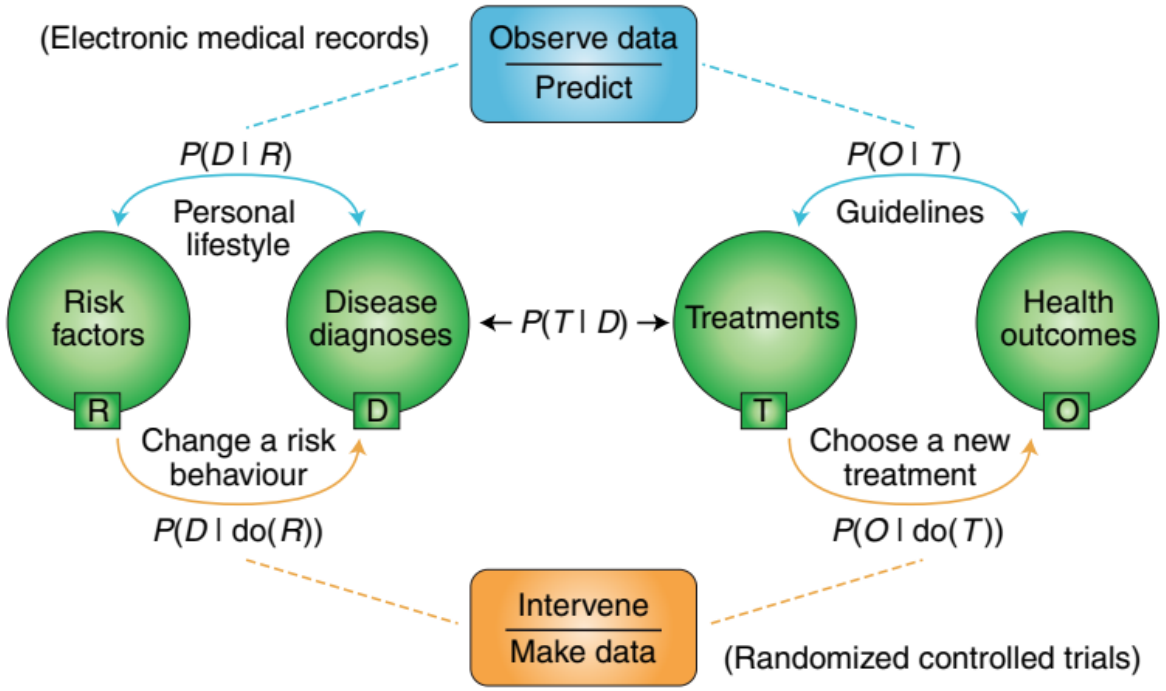
What we can compare

Treatment choices follow severity, access, and preference

This holds when we analyze past care, and again when AI begins to shape future care.

Wang, Li, Lin, Guo. npj Digit Med 2026;9:424; Wang, Smith, Lipori, Sun, Shenkman, Liu, Wu, Guo. npj Health Syst 2026;3:83

Observing is not intervening



P(Y | X) ≠ P(Y | do(X))

Nicotine-stained teeth predict lung cancer, but cleaning teeth does not prevent it

- Observed data give conditional probabilities; decisions need interventional ones
- **They coincide under randomization, or when exchangeability and other identification conditions hold**

Prosperi, Guo et al. Nat Mach Intell 2020; example adapted from M. Prosperi, Causal Health AI presentation

What AI adds to causal inference, and what it cannot replace

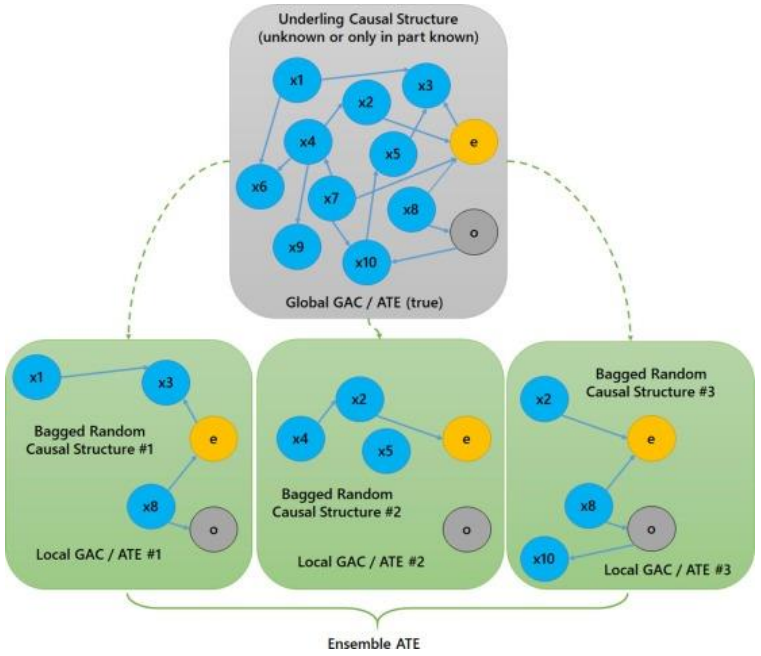


Figure: Ghosh, Guo, Prosperi. JAMIA 2021; Prosperi, Guo et al. JBI 2021

AI ADDS

Measurement of exposures, confounders, and outcomes from notes, images, multimodal records

Estimation with flexible models for high-dimensional confounding and effect heterogeneity

Decision support that turns estimates into recommendations

STILL REQUIRED

A well-defined causal question and estimand

A study design the data can support

Explicit identifying assumptions and appropriate diagnostics

More data and larger models do not substitute for these

Target trial emulation: specify the trial you would run

1 **Eligibility criteria**	**2** **Treatment strategies**	**3** **Treatment assignment**	**4** **Start and end of follow-up**
5 **Outcomes**	**6** **Causal contrasts**	**7** **Analysis plan**	**Specify the protocol before outcome-driven analysis decisions**

A well-designed emulation helps prevent avoidable biases such as immortal time. Regulators increasingly accept it as a structured approach to real-world evidence when trials are not feasible.

Hernán, Wang, Leaf. JAMA 2022

The trial we want is not always the trial EHR data can support

Fundamental problem	Where it arises in EHR data
Who is observed?	Care-seeking selection, eligibility, time zero, loss to follow-up
What is measured?	Treatment exposure, outcomes, confounders, observation intensity
What can be compared?	Confounding, positivity, treatment strategies, transportability

Each problem can change the question being answered, not only the precision of the answer.

Synthesized from Table 1, Wang, Li, Lin, Guo. npj Digit Med 2026;9:424 (full table in backup)

Who is observed, and what is measured

Disease onset
unobserved

→

First encounter
cohort entry

→

Treatment decision
target trial time zero

→

Recorded diagnosis or Rx
EHR time zero

Who is observed

Entry depends on care seeking; events before the first visit are unseen (left truncation)

Conditioning on engagement can open collider paths; anchoring time zero to a record can introduce immortal time

What is measured

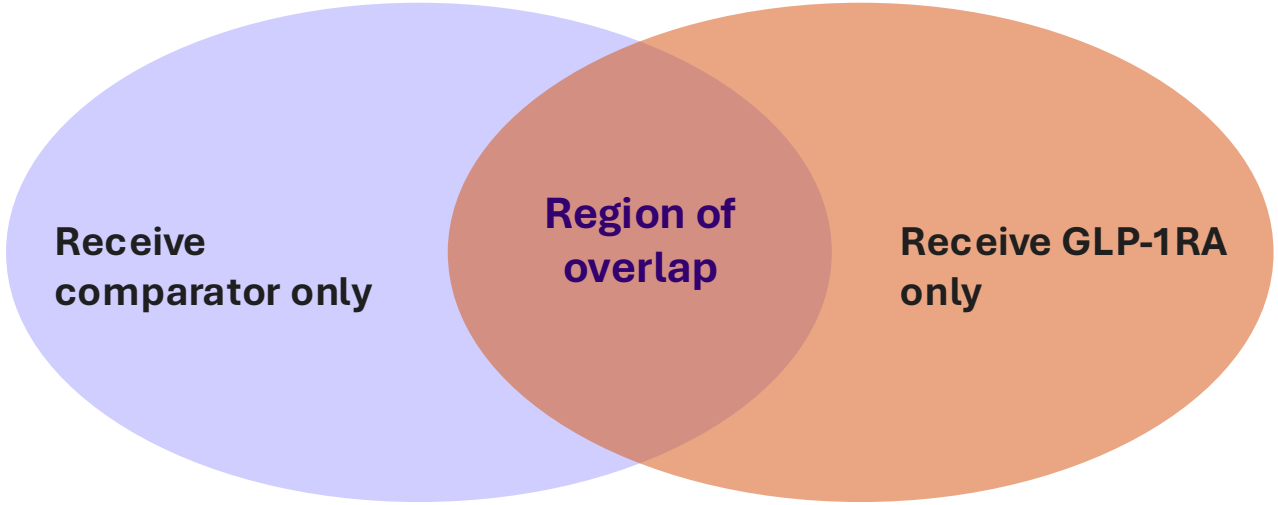
Prescriptions record prescribing, not necessarily use; outcomes are detected through care, unevenly by severity and monitoring

NLP and LLMs recover eligibility and outcomes from notes, with misclassification (e.g., Lung-RADS, F1 = 0.89)

Open question: what happens to a causal estimate when the exposure, a confounder, or the outcome comes from NLP?

Wang, Li, Lin, Guo. npj Digit Med 2026;9:424; Yang, Wu, Guo. JHIR 2024; Yang, Liang, Guo. AMIA 2025

Who can be compared: overlap shapes the question



Patient characteristics that drive treatment choice

- **Structural non-overlap:** the contrast is not identifiable for those patients from the observed data without additional assumptions
- **Weak overlap:** identifiable in principle, but unstable and model-dependent
- **Trimming or overlap weighting** stabilizes estimates by changing the target population
- **Report it as a restricted estimand**

Wang, Li, Lin, Guo. npj Digit Med 2026;9:424

From the intended estimand to the data-defined estimand

Clinical causal question

Intended target population and treatment contrast

EHR operational constraints

Observability, measurement, timing, treatment overlap

Data-defined estimand

The contrast the available data can support, if assumptions hold

Statistical adjustment cannot recover a causal contrast that the data do not support. If constraints rule out the intended estimand, redefine and name it explicitly.

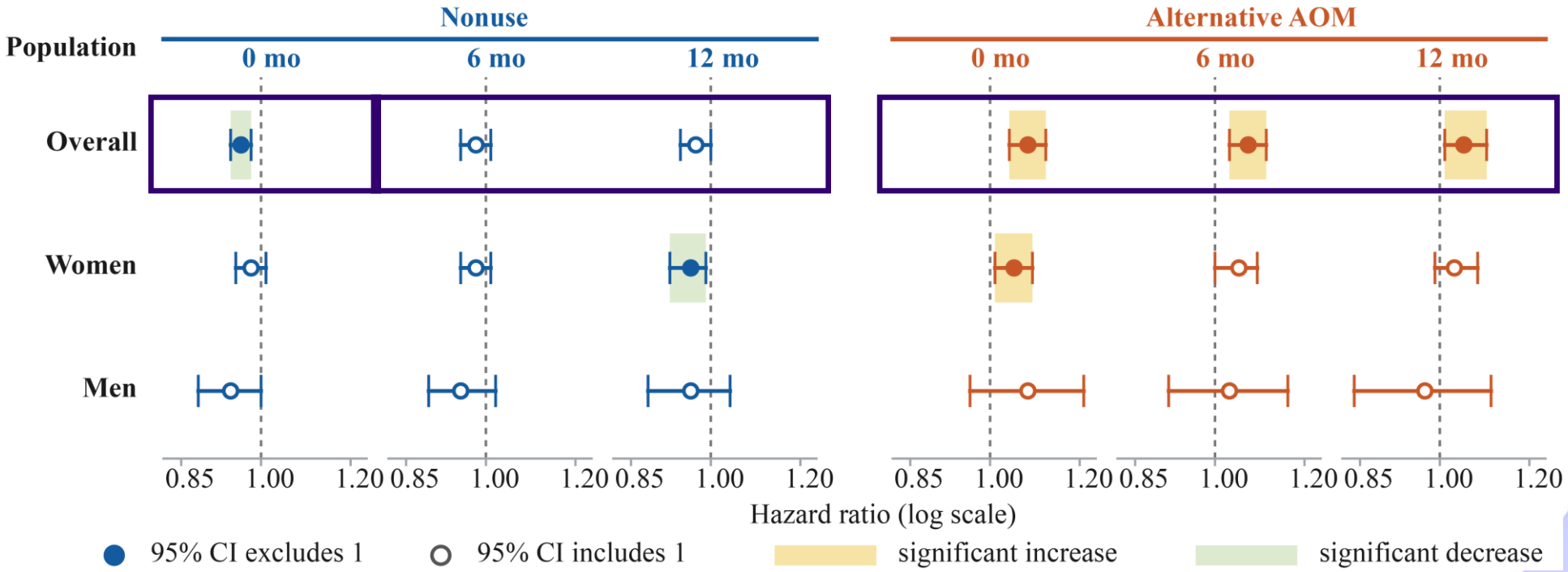
Well suited when baseline, outcomes, and comparable treated patients are all observable

Not identifiable as posed when it hinges on unobserved onset or adherence: reframe around an observable decision, or do not emulate

Wang, Li, Lin, Guo. npj Digit Med 2026;9:424

How study design changes the estimated association

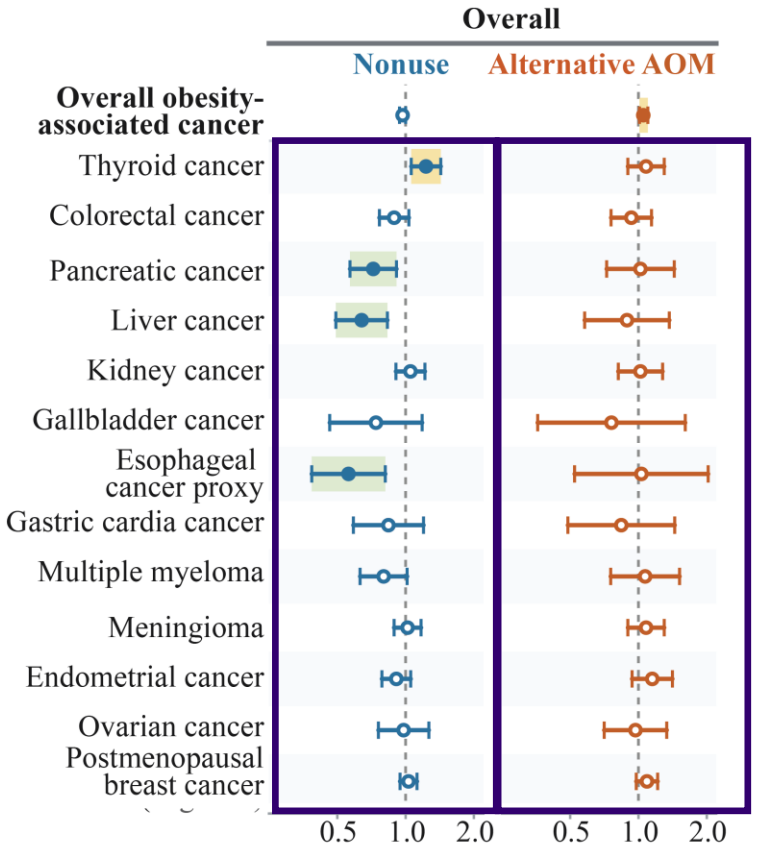
GLP-1RA initiation and obesity-associated cancer, adults with obesity and no type 2 diabetes (Epic Cosmos target trial emulation; 580,511 matched pairs in the primary analysis)



Wang, Sheer, Li, et al., Guo. Manuscript under review

Under review: please do not cite

Site-level signals depend on the comparator



Nonuse: four sites cross the Bonferroni-adjusted threshold

Thyroid higher (HR 1.23); pancreatic (0.72), liver (0.64) and esophageal adenocarcinoma proxy (0.56) lower

Active comparator: no site remains significant

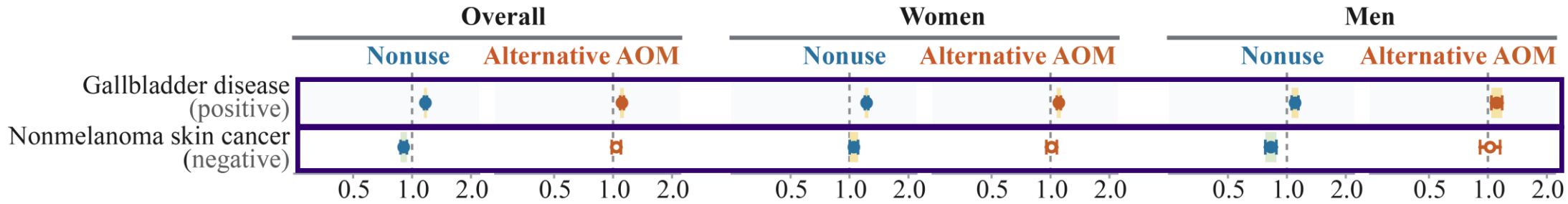
Intervals are wider; loss of significance does not show that the associations are absent

12-month landmark; Bonferroni-adjusted simultaneous CIs for sites

Wang, Sheer, Li, et al., Guo. Manuscript under review

Under review: please do not cite

Positive and negative control outcomes



12-month landmark cohorts; nonuse and alternative AOM comparators

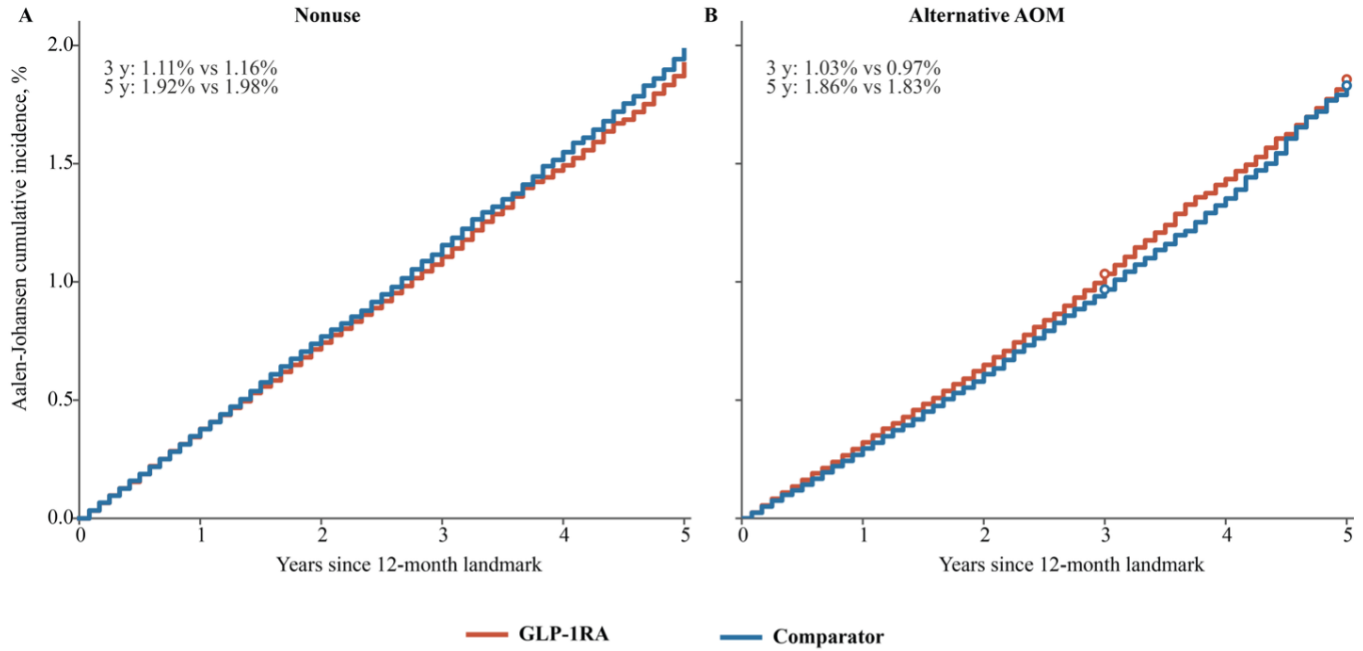
Positive control, gallbladder disease: reproduces the established association in both designs and every population.

Negative control, nonmelanoma skin cancer: near null with the active comparator in every population; off-null with nonuse, in opposite directions (HR 0.91 overall, 0.83 in men, 1.05 in women), raising concern about residual confounding or differential detection.

Wang, Sheer, Li, et al., Guo. Manuscript under review

Under review: please do not cite

Clinical implications of the GLP-1RA findings



Cumulative incidence of obesity-associated cancer after the 12-month landmark

< 1 per 1,000

observed absolute difference at 3 and 5 years, either comparator

For the clinical decision

Small observed absolute differences in obesity-associated cancer risk over available follow-up

Does not support prescribing GLP-1RAs for cancer prevention; weigh cancer risk alongside established benefits, other risks, and remaining uncertainty

- Limits: median follow-up about 3 years for long-latency cancers; code-based outcomes; residual confounding and detection cannot be excluded

Wang, Sheer, Li, et al., Guo. Manuscript under review

Under review: please do not cite

From average effects to individualized and sequential decisions

Does treatment A work on average?	Causal effect estimation, target trial emulation
For whom does it work better?	Heterogeneous treatment effects
What should we do next as the patient changes?	Dynamic treatment strategies, policy learning
Does the recommendation stay valid?	Decision-specific validation, feedback monitoring

Clinical decisions change both patient outcomes and the data available for future decisions.

Synthesis

Connecting target trial emulation and clinical digital twins

Target trial emulation

Learns from data shaped by past care decisions

Key issue: intended vs. data-defined estimand



Intelligent clinical digital twins

Shape the care decisions that generate their future data

Key issue: predictive vs. decision-support validity

Clinical decisions shape the data used for causal inference. AI-supported decisions can also shape the data used for future predictions and recommendations.

Wang, Li, Lin, Guo. npj Digit Med 2026;9:424; Wang, Smith, Lipori, Sun, Shenkman, Liu, Wu, Guo. npj Health Syst 2026;3:83

When AI acts in care, it joins the data-generating process

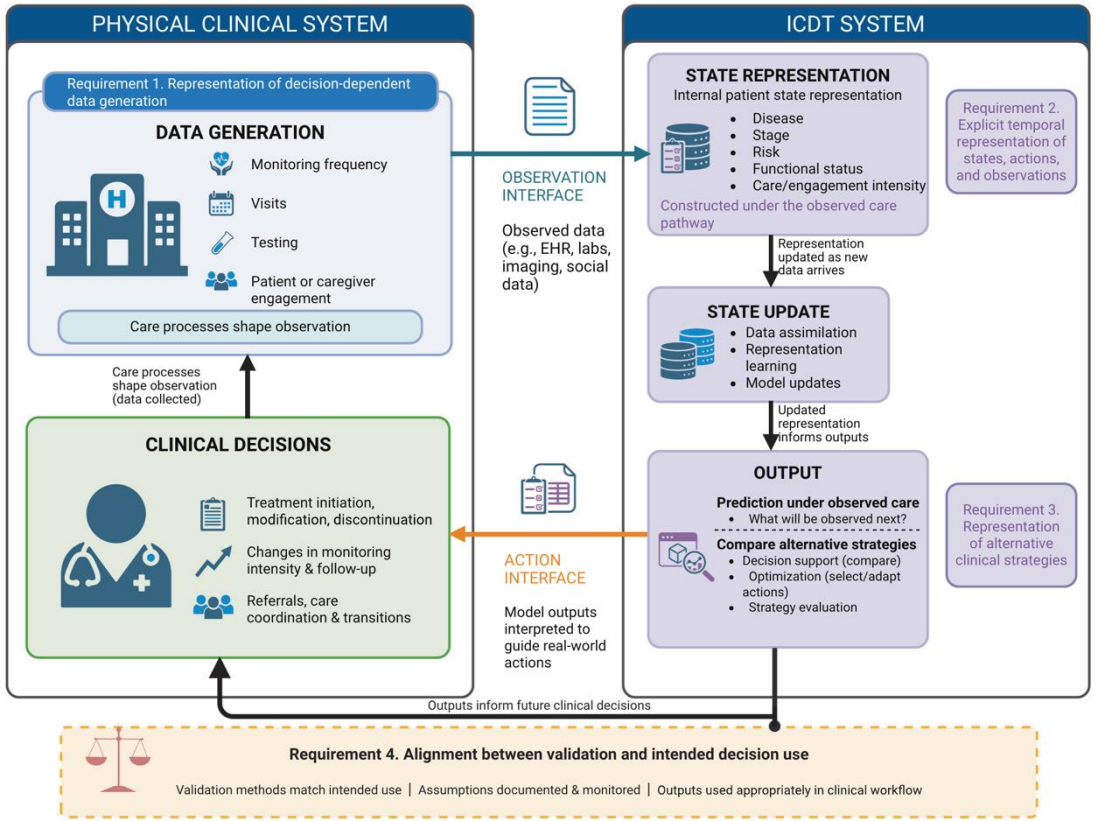


Figure 1, Wang, Smith, Lipori, Sun, Shenkman, Liu, Wu, Guo. npj Health Syst 2026;3:83 (BioRender)

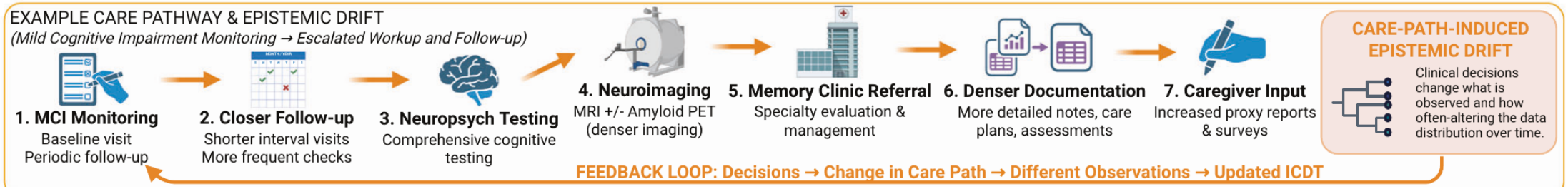
Prediction under observed care

What will be observed next if care continues as before? Validated by discrimination and calibration

Decision support

What would happen under strategies not taken? Needs explicit strategies, support, assumptions, and claim-specific validation

When AI changes care, it changes the data



- A risk alert for mild cognitive impairment prompts closer follow-up, testing, imaging, and a memory clinic referral
- Documentation grows denser; reports shift from patient to caregiver
- The record can look like faster decline, partly because of how the patient was observed

Comparing monitoring vs. referral from this updated state

is distorted if the state is treated as a neutral baseline and the care history that produced it is ignored

Conceptual scenario, Figure 2, Wang, Smith, Lipori, Sun, Shenkman, Liu, Wu, Guo. npj Health Syst 2026;3:83

Epistemic drift: a proposed mechanism with measurable signals

A model can remain well calibrated while the meaning of its patient representation shifts, from disease alone toward care intensity, prior alerts, and reporting regimes.

Visit frequency	**New specialty referral**	**More testing and imaging**	**Documentation density**	**Caregiver or proxy reporting**

Indicators to monitor. Standard validation may not detect drift, because accuracy can stay stable.

Wang, Smith, Lipori, Sun, Shenkman, Liu, Wu, Guo. npj Health Syst 2026;3:83

Four structural requirements for decision-oriented clinical AI

1 Represent decision-dependent data generation

Separate change in disease from change in observation and documentation

2 Represent states, actions, and observations in time

Preserve clinical order: what preceded, triggered, or resulted from a decision

3 Represent alternative clinical strategies

Keep observed-care tracking separate from strategy evaluation (target trial emulation, g-formula)

4 Align validation with intended decision use

Match the evidence to the specific decision claim

Adapted from Table 1, Wang, Smith, Lipori, Sun, Shenkman, Liu, Wu, Guo. npj Health Syst 2026;3:83; requirement 3 links back to target trial emulation

Validation should match the intended clinical use

Claim	Evidence needed
Forecasts outcomes under current care	Predictive validity: discrimination, calibration, uncertainty on the observed care pathway
Compares clinical strategies	Causal identification: defined strategies, support diagnostics, estimands with stated assumptions
Stays valid as it guides care	Ongoing monitoring: decision-use audit, calibration by care pathway, drift
Improves patient outcomes	**Clinical impact: prospective evaluation of clinical use and patient outcomes**

Identifying an effect is not proof of benefit. The risk is not a lack of validation; it is a mismatch between what was validated and how outputs are used.

Adapted from Table 1, Wang, Smith, Lipori, Sun, Shenkman, Liu, Wu, Guo. npj Health Syst 2026;3:83

A research agenda for causal AI in clinical care

1

Identify effects from imperfect, decision-dependent data

AI-derived and misclassified measurements; care-driven entry and observation; EHR-aware trial emulation; identification before estimation

2

Estimate strategies under irregular, informative observation

Longitudinal patient representations; dynamic treatment strategies; heterogeneity; transportability; uncertainty

3

Validate adaptive systems when deployment changes the data

Feedback-aware design; claim-specific validation; drift monitoring; evaluation of clinical impact

Evidence informs decisions; it does not make them. Actionability, feasibility, fairness, uncertainty, and patient preferences remain part of every decision, and need clinicians, patients, and communities at the table.

Fairness example: Angell, Li, et al., Guo. MedInfo 2025; responsible AI framing adapted from M. Prosperi

Three take-home messages

1 Start with the decision, not the model

The causal question and estimand should determine the design and the analysis

2 Clinical decisions shape what the data can support

Clinical data record care decisions; design choices can reverse an answer

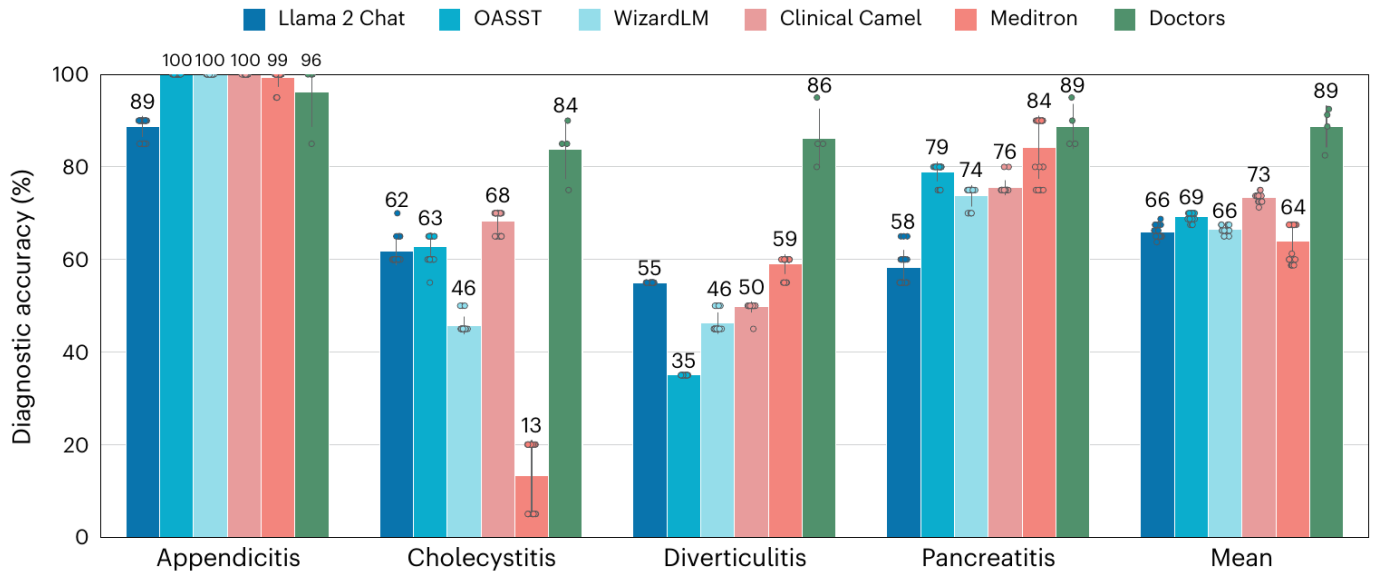
3 As AI changes care, re-evaluate the evidence supporting its decisions

As AI joins care, it reshapes its own data; monitor and re-validate for the claim you make

Thank you · yiguo@ufl.edu

Backup: strong exam performance has not yet become better decisions

Benchmark: LLMs vs. physicians on real MIMIC cases



Diagnostic accuracy, 4 abdominal conditions, subset of 80 of 2,400 cases (Hager et al.)

Randomized trial: 50 physicians

76% vs. 74%

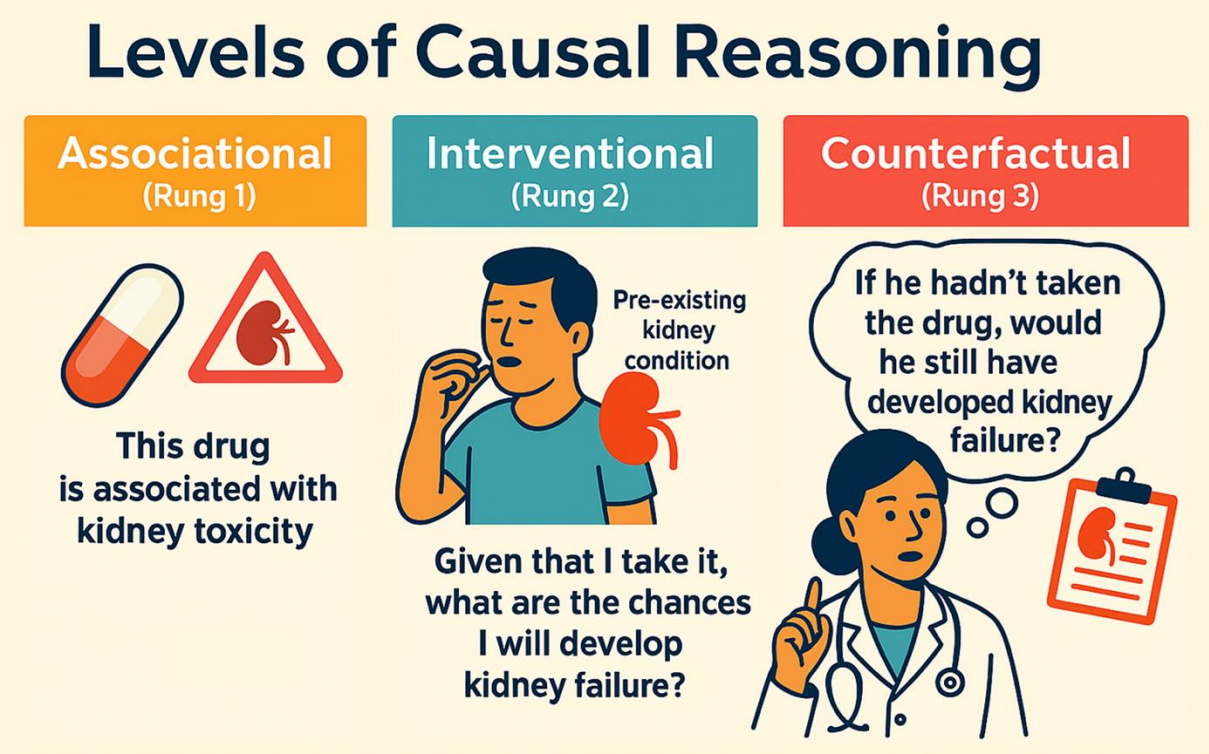
Diagnostic reasoning with vs. without GPT-4 access (P = .60)

+16 points

GPT-4 alone vs. physicians using conventional resources (secondary analysis)

Hager et al. Nat Med 2024;30:2613–22 (figure reproduced); Goh et al. JAMA Netw Open 2024;7:e2440969

Backup: clinical questions climb the causal ladder



Prescription *Intervention, counterfactual*

What if an alternative treatment is chosen?

Prognosis *Association under observed care*

What is the probability of survival under standard of care?

Diagnosis *Association*

If the person has these symptoms, what is the disease?

Figure and framing adapted from M. Prosperi, Causal Health AI presentation

Backup: design components and EHR constraints in detail

Target trial component	EHR data constraint	Implication for identifiability
Eligibility criteria	Cohort entry driven by healthcare use; disease onset unobserved; left truncation	Without selection or transportability assumptions, estimand pertains to the care-seeking population
Baseline (time zero)	Anchored to diagnosis or encounter rather than decision time	Residual misalignment persists; estimand may shift
Treatment strategies and assignment	Treatment is dynamic; exposure incompletely recorded	Incomplete longitudinal exposure data may prevent identification of sustained-strategy effects
Follow-up and censoring	Informative loss to follow-up; competing events	Risk sets altered; contrast differs from target trial
Outcome measurement	Differential detection; delayed or heterogeneous recording	Observed outcome differs from trial-defined outcome
Causal contrast and estimand	Residual confounding; structural non-overlap	Target estimand not identifiable; data-defined estimand
Analysis plan	Irregular measurement; multi-site heterogeneity	Analytic sophistication cannot recover unsupported questions

Adapted from Table 1, Wang, Li, Lin, Guo. npj Digit Med 2026;9:424

Backup: site-specific and sex-specific estimates

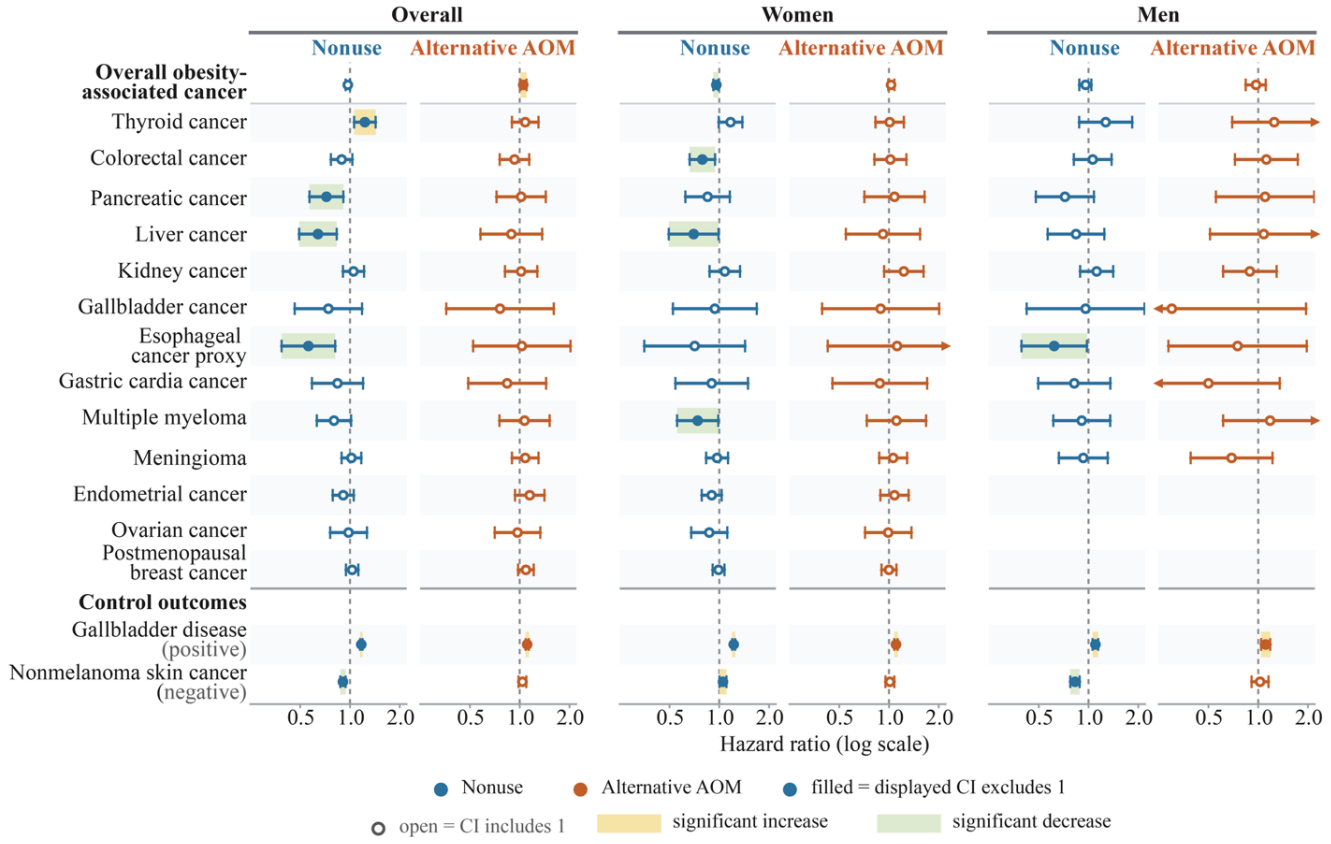


Figure 1, Wang, Sheer, Li, et al., Guo. Manuscript under review; site-specific CIs Bonferroni-adjusted

Under review: please do not cite